Evaluating LLM-Driven Knowledge Graph Construction: A Multi-Metric Approach Addressing Hallucination and Omission

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Abstract. Recent advancements in large language models (LLMs) have shown significant potential in automating knowledge graph (KG) construction from unstructured text. This paper introduces an enhanced evaluation framework for KG construction, incorporating metrics for hallucination, omission, and graph similarity. We propose a practical threshold of 95% for graph matching using BERTScore. Our experiments focus on the Mistral model, comparing its original and fine-tuned versions in zero-shot and few-shot settings. Results show that fine-tuned models significantly improve KG construction accuracy while reducing hallucination and omission. However, we also observe that fine-tuning may impact negatively the generalization capabilities on diverse datasets.

Keywords: Text-to-Knowledge Graph, Large Language Models, Zero-Shot Prompting, Few-Shot Prompting, Fine-Tuning, Hallucination

1 Introduction

Knowledge Graphs (KGs) play a crucial role in organizing complex information across diverse domains [1]. The challenge of constructing KGs from unstructured textual data remains significant. Large Language Models (LLMs) have shown promising results in this area, leveraging techniques such as in-context learning and prompt design [2, 3].

Our study builds upon previous work evaluating various models using metrics like precision, recall, F1 score, triple matching, and graph matching. We introduce a refined approach to address critical issues of hallucination and omission in KG construction.

2 Background

Recent research has explored three main approaches for KG construction using LLMs. The first approach, Zero-Shot Prompting (ZSP), enables KG construction without task-specific training data [4, 5]. The second approach, Few-Shot Prompting (FSP), focuses on constructing KGs with limited training examples [6]. The third approach, Fine-Tuning (FT), involves adapting pre-trained language models to specific knowledge domains [7].

Evaluation metrics for KG construction typically include semantic consistency (hallucination, omission, accuracy) and linguistic coherence [8]. Recent studies have introduced more sophisticated metrics such as Graph BERTScore (G-BS) and Graph Edit Distance (GED) [6, 9].

3 Methodology

We propose an enhanced evaluation framework incorporating several key components (Fig 1). First, we use BERTScore for graph similarity, setting a 95% threshold for graph matching. Second, we introduce refined hallucination and omission metrics using Optimal Edit Paths (OEP). Finally, we present a new metric: Graph Matching using Graph BERTScore (GM-GBS). Our experiments focus on the Mistral model [10], comparing its original and fine-tuned versions in zero-shot and few-shot settings. We use the WebNLG+2020 [11] and KELM-sub [6] datasets for evaluation.

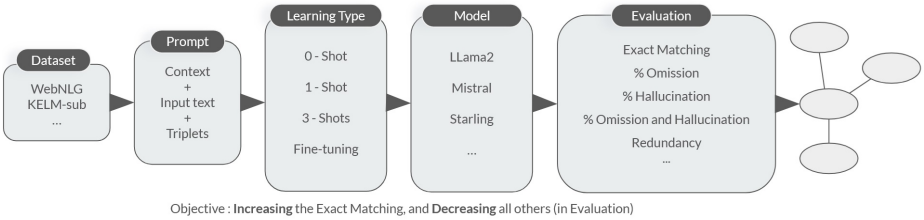

Fig. 1. Overall experimentation's process

4 Experiments and Results

We evaluated the performance of both original and fine-tuned Mistral models using Zero-Shot Prompting (ZSP) and Few-Shot Prompting (FSP). Our analysis incorporated several metrics, including G-F1, T-F1, G-BS, GED, hallucination, and omission rates.

Key findings of the study are as follows. The first part of Table 1 shows that he fine-tuned Mistral outperformed the original model, particularly in Few-Shot scenarios. The fine-tuned Mistral with seven shots achieved superior performance across most metrics. However, the second part of the table shows that fine-tuning on domain-specific data (e.g., WebNLG) can negatively impact the model's generalization capabilities on diverse datasets like KELM-sub. Additionally, the GM-GBS metric provided valuable insights into semantic similarity between generated and ground truth graphs with 95% as threshold of BertScore.

Table 1. Comparison of performance metrics and models on WebNLG test dataset. Lower values indicate better performance for GED, Hall., and Omis.

Model — Metric	G-F1	T-F1	G-BS	GED	Hall.	Omis.	GM-GBS
Mistral-0	2.30	3.27	77.87	15.84	20.35	31.31	33.27
Mistral-7	18.72	28.44	87.54	10.13	17.88	21.14	51.88
Mistral-FT-0	31.93	44.08	86.89	8.25	**13.55**	18.27	54.97
Mistral-FT-7	**34.68**	**49.11**	**91.99**	**6.69**	14.90	14.39	57.72
Mistral-6 (KELM-sub)	7.59	12.45	81.23	16.29	61.16	**7.64**	26.86
Mistral-FT-6 (KELM-sub)	31.37	47.49	91.27	7.51	27.37	8.26	**58.40**
Mistral-7	5.50	11.35	81.77	13.74	6.72	61.09	28.66
Mistral-FT-0	2.17	8.55	78.29	14.35	7.22	56.28	12.88
Mistral-FT-7	2.89	9.92	78.42	13.63	**6.22**	61.00	13.66
Mistral-6 (KELM-sub)	**12.00**	**31.08**	**85.49**	**10.82**	25.50	**32.44**	**38.88**
Mistral-FT-6 (KELM-sub)	4.00	17.66	84.30	12.50	11.06	4817	36.22

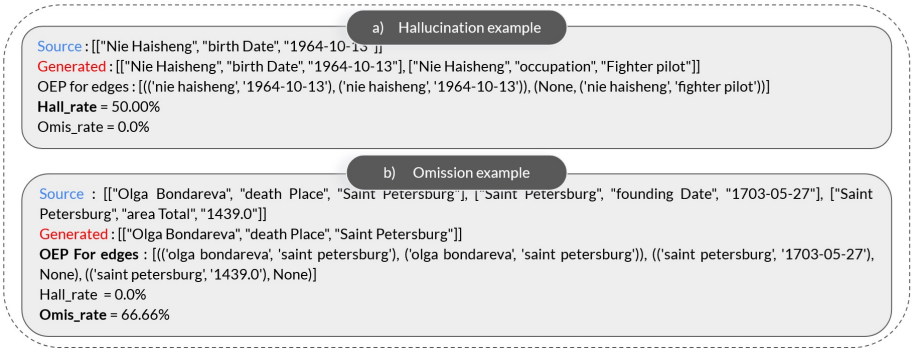

Fig. 2. Results examples

As illustrated in Fig 2, our metric precisely calculates the percentage of hallucinations and omissions in the generated graphs at the triple level. For example, if a generated graph contains 2 triples and 1 of them are not present in the ground truth graph, the hallucination rate is approximately 50%. Similarly, for omissions, if the generated graph is missing some triples present in the ground truth graph, the omission rate is calculated accordingly.

5 Conclusion and Future Work

Our study demonstrates the effectiveness of fine-tuned LLMs in KG construction tasks, while also highlighting the challenges in maintaining broad generalization capabilities. Future research directions include refining evaluation metrics to

account for synonyms in generated graphs, leveraging LLMs for data augmentation in T2KG construction, and expanding evaluations to a broader range of domains and datasets. Furthermore, combining automated metrics with human evaluation could offer richer quality assessment.

These directions will contribute to advancing the field of T2KG construction and enhancing the capabilities of language models in producing accurate and contextually appropriate knowledge graphs.

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